(a) Methodological Ground Simulation & Chamber Parity

Chamber 1: GMF Control

Magnetic Regime: 45–50 μT Ambient GMF

Control Purpose: Terrestrial baseline

Chamber 2: Near-Null (NNMF)

Magnetic Regime: <10 nT (Shielded/Cancelled)

Control Purpose: Lunar simulation

Chamber 3: Sham Control

Magnetic Regime: Current matched; B = 0

Control Purpose: Controls ohmic heat & EMI

(b) Active Causal Rescue Validation Protocol

Causal Rescue Inside Magnetic Shielding

Step 1: Hypomagnetic Exposure

<10 nT inside μ-metal shield

Elicits germination & pigment phenotype

Step 2: Vector Re-application

45 μT active vector restored

Re-applied via triaxial Helmholtz coils

Step 3: Biological Reversal

Phenotype rescued to GMF level

Formally proves magnetic causality

Rules out microclimatic, volatile, or enclosure artifacts

(c) Predeclared Flight Decision Framework for Lunar Agricultural Payloads (LEAF / BLSS)

GREEN GATE: Nominal (Retire Risk)

Empirical Thresholds:

- Germination rate drop < 5%
- Emergence delay (Δt50) < 6 h
- Fv/Fm photochemical yield nominal
- No significant biomass penalty

Flight Operational Action:

Retire hypomagnetic risk from critical path.
Proceed with standard autonomous operations, ambient LED photoperiods, & baseline harvest.

AMBER GATE: Operational Shift

Empirical Thresholds:

- Germination rate drop 5–20%
- Emergence delay (Δt50) 6–24 h
- 10–25% reduction in seedling biomass
- Moderate chlorosis / delayed greening

Flight Operational Action:

Adjust mission schedule:
- Increase seed sowing density by 15–20%
- Extend growth duration prior to harvest
- Boost blue-light ratio for photomorphogenesis.

RED GATE: Hazard Mitigation

Empirical Thresholds:

- Germination capacity drop > 20%
- Severe emergence failure (Δt50 > 24 h)
- Severe ROS leak & photosynthetic failure
- > 30% collapse in harvestable yield

Flight Operational Action:

Mandate hardware & biological mitigation:
- Deploy low-power active bias coils (40–50 μT)
- Switch to stress-tolerant crop cultivars
- Supplement feed with antioxidant cofactors.